KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

SAP Procure-to-Pay (P2P) Cycle

Full Purchasing Cycle — SAP MM & FI

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1 Problem Statement

Many mid-sized manufacturing companies in India still manage procurement through spreadsheets, email chains, and phone calls. While this works at a small scale, it breaks down quickly as the business grows. **TechNova Industries Pvt. Ltd.**, a fictional electronics manufacturer based in Bhubaneswar, faced exactly these challenges:

Problem	Business Impact
No centralized purchase requisition system	Delays of 3–5 days per approval
Informal vendor selection (no comparison)	Higher procurement costs
Paper-based goods receipt	Lost records, delivery disputes
Manual invoice matching	Duplicate and incorrect payments
No payment schedule tracking	Vendor relationship issues

This project implements the full SAP Procure-to-Pay (P2P) cycle to solve these problems using SAP MM and SAP FI modules.

2 Objectives

- Understand the complete Procure-to-Pay business process from a practical perspective
- Execute each SAP transaction step using the correct T-Code
- Demonstrate how SAP MM and SAP FI modules integrate in a real procurement workflow
- Show how 3-way matching (PO + GR + Invoice) prevents financial errors
- Document the entire process clearly for academic and practical reference
- Present a consistent real-world scenario throughout all steps

3 Solution Description

The solution is based on the standard SAP P2P process, configured and executed in an SAP ECC 6.0 training sandbox. The business scenario used throughout this project is:

📦 **Project Scenario — TechNova Industries**

Company: TechNova Industries Pvt. Ltd. (Company Code: TN01)
Requirement: Purchase 100 units of ATmega328P microcontroller chips
Vendor Selected: ChipWorld India (after comparing 3 vendor quotes)
Total Value: ₹13,800 | Payment Terms: Net 30 Days

This single scenario is used across all 7 steps to show a complete, connected document flow from Purchase Requisition to Vendor Payment.

4 SAP P2P Process — Steps 1 to 4

Step 1 ME51N **Purchase Requisition (PR)**

The warehouse team at TechNova identifies that microcontroller chip stock is running low. They raise a PR in SAP for **100 units of CHIP-ATM-328P** required by 15.04.2026. The PR is an internal document — it goes to the Purchasing team, not to any vendor.

Output: PR Number 1000012345 created | Routed to Purchasing Department

Step 2 ME41 **Request for Quotation (RFQ)**

The Purchasing team creates RFQ documents and sends them to three approved vendors requesting prices and delivery terms for the 100 chips. Each vendor is given until 05.04.2026 to respond.

Output: RFQ# 6000001 (MicroElec), 6000002 (ChipWorld India), 6000003 (TechParts Hub)

Step 3 ME49 **Vendor Comparison & Selection**

SAP generates a comparison report showing all three vendor quotes side by side. ChipWorld India offers the lowest price and fastest delivery.

Vendor	Unit Price	Total (100 units)	Delivery	Decision
MicroElec Supplies	₹150	₹15,000	7 days	❌ Rejected
ChipWorld India	**₹138**	**₹13,800**	**5 days**	✅ Selected
TechParts Hub	₹145	₹14,500	6 days	❌ Rejected

Output: ChipWorld India selected | Savings of ₹1,200 vs next cheapest vendor

Step 4 ME21N **Purchase Order (PO)**

A formal, legally binding Purchase Order is created for ChipWorld India. The PO references the PR and accepted RFQ. It specifies the agreed price, delivery date, and payment terms.

Output: PO# 4500001234 | Vendor: ChipWorld India | ₹13,800 | Delivery by 12.04.2026 | Terms: Net 30

5 SAP P2P Process — Steps 5 to 7

Step 5 MIGO **Goods Receipt (GR)**

On 11.04.2026 the delivery from ChipWorld India arrives. The warehouse team opens MIGO, enters PO# 4500001234, and confirms receipt of all 100 units with movement type 101. SAP updates inventory and posts an accounting entry.

Accounting Entry: Dr. Inventory Account (Raw Material) → Cr. GR/IR Clearing Account

Output: Material Document 5000001234 | Stock: +100 units in SL01 | Inventory value updated in FI

Step 6 MIRO **Invoice Verification (3-Way Match)**

ChipWorld India sends an invoice for ₹13,800. The Finance team enters it in MIRO. SAP automatically runs a 3-way match:

Document	Quantity	Amount	Status
Purchase Order (PO)	100 EA	₹13,800	✅ Match
Goods Receipt (GR)	100 EA	—	✅ Match
Vendor Invoice	100 EA	₹13,800	✅ Match

Accounting Entry: Dr. GR/IR Clearing Account → Cr. Vendor Account (ChipWorld India) ₹13,800

Output: Invoice Document 5105000123 | Vendor payable ₹13,800 | Status: Ready for Payment

Step 7 F110 **Automatic Payment Run**

On 11.05.2026 (30 days after invoice posting), the Finance manager runs F110. SAP identifies all invoices due, generates a bank transfer file, and clears the vendor's open item automatically.

Accounting Entry: Dr. Vendor Account (ChipWorld India) → Cr. Bank Account ₹13,800

Output: Payment Document 1400000456 | ₹13,800 paid to ChipWorld India | Vendor account fully cleared

6 Complete Document Flow

Purchase Requisition ME51N | PR# 1000012345	Warehouse raises internal purchase request
↓	
Request for Quotation ME41 | RFQ# 6000001-3	Sent to 3 vendors; ChipWorld India wins comparison
↓	
Purchase Order ME21N | PO# 4500001234	Formal PO issued to ChipWorld India for ₹13,800
↓	
Goods Receipt MIGO | MD# 5000001234	100 units received; stock updated in SL01
↓	
Invoice Verification MIRO | INV# 5105000123	3-way match passes; invoice approved for payment
↓	
Vendor Payment F110 | PAY# 1400000456	₹13,800 paid via bank transfer; account cleared

7 Technology Stack

ERP SYSTEM	PRIMARY MODULE	SECONDARY MODULE
SAP ECC 6.0 (Sandbox)	SAP MM — Materials Mgmt	SAP FI — Financial Acctg
SAP GUI VERSION	**COMPANY CODE**	**DOCUMENTATION**
SAP GUI 7.60	TN01 (TechNova Ind.)	Markdown + GitHub

8 Screenshots Documentation

Screenshots are stored in the `/screenshots` folder as per the naming convention.

#	T-Code	Screenshot File	Key Content
1	ME51N	01_ME51N_PR_creation.png	PR creation with material & quantity
2	ME41	02_ME41_RFQ_creation.png	RFQ with vendor list and deadline
3	ME49	03_ME49_vendor_comparison.png	Price comparison table — all 3 vendors
4	ME21N	04_ME21N_PO_creation.png	PO with vendor, price, and delivery date
5	MIGO	05_MIGO_goods_receipt.png	GR with movement type 101, qty confirmed
6	MIRO	06_MIRO_invoice_verification.png	3-way match with Balance = 0
7	F110	07_F110_payment_run.png	Payment run log, cleared vendor item
8	Any	08_document_flow.png	Full document chain PR → Payment

9 Future Improvements

- **Approval Workflow:** Add PR and PO approval routing using SAP Business Workflow
- **Vendor Portal:** Integrate SAP SRM so vendors submit quotes through an online portal
- **Analytics Dashboards:** Build procurement KPI dashboards in SAP Analytics Cloud
- **Partial Delivery Scenarios:** Demonstrate split goods receipts and partial payments
- **S/4HANA Migration:** Run the same process using S/4HANA and Fiori-based apps

10 Conclusion

This project successfully demonstrates the complete Procure-to-Pay (P2P) cycle in SAP across seven structured steps — from Purchase Requisition (ME51N) through to Automatic Payment (F110). Using TechNova Industries as a consistent business context, every step was shown with realistic data, document numbers, and accounting entries.

The most valuable insight from this project is the power of **3-way matching in MIRO**. By verifying that the PO, GR, and invoice are all consistent before approving payment, SAP eliminates the most common source of procurement-related financial loss — paying for goods that were never ordered or never received.

✅ **Project Completion Summary**

7 SAP steps executed | 2 modules integrated (MM + FI) | 1 complete document flow | ₹13,800 procurement cycle demonstrated end-to-end